

- (a) Die uitwyking y van 'n golf, wat met 'n snelheid v in die positiewe x -as rigting beweeg, word op tyd $t = 0$ gegee deur:

The deviation y of a wave travelling in the positive x -direction with velocity v , is given at $t = 0$ as:

$$y(x, 0) = A \cos\left(\frac{2\pi x}{\lambda}\right)$$

- (i) Toon aan dat op enige tyd t die golf funksie geskryf kan word as:
Show that at any instant this wave function can be written as:

$$y(x, t) = A \cos(kx - \omega t)$$

$$\begin{aligned} y(x, t) &= A \cos\left[\frac{2\pi}{\lambda}(x - vt)\right] \\ &= A \cos\left[\frac{2\pi x}{\lambda} - \frac{2\pi}{\lambda}(vt)\right] \\ &= A \cos\left[kx - \frac{2\pi}{\lambda}(\lambda f)t\right] \\ &= A \cos[kx - 2\pi f t] \\ &= A \cos[kx - \omega t] \end{aligned}$$

(3)

- (ii) Skryf uitdrukkings neer vir die golfgetal k en die hoeksnelheid ω en gee hulle eenhede.

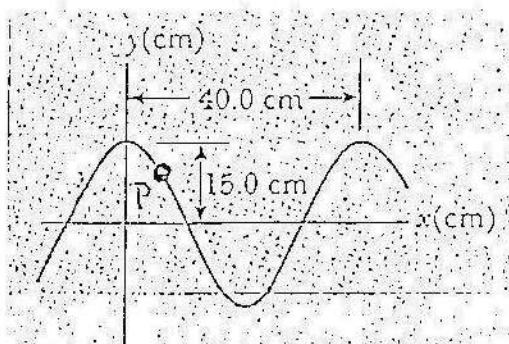
Give expressions for the wave number k and the angular velocity ω . Also give their units.

$$\begin{aligned} k &= \frac{2\pi}{\lambda} \quad \checkmark \quad \text{rad/m} \quad \checkmark \\ \omega &= \frac{2\pi}{T} \quad \checkmark \quad \text{rad/s} \quad \checkmark \end{aligned}$$

(4)

- (b) 'n Golf $y = A \sin(kx - \omega t + \delta)$ beweeg met 'n snelheid v in die positiewe x -as rigting. Die amplitude van die golf is 15 cm, sy golflengte is 40 cm en sy frekwensie is 8 Hz. Met $t = 0$ en $x = 0$ is die vertikale verplasing $y = 15$ cm.

A wave $y = A \sin(kx - \omega t + \delta)$ is travelling in the positive x -direction with a velocity v . It has an amplitude of 15 cm, a wavelength of 40 cm and a frequency of 8 Hz. The vertical displacement at $t = 0$ and $x = 0$ is $y = 15$ cm.



Vraag / Question 1(b)(i)(Vervolg)(Continue) /

- (i) Bepaal die golflengte λ en gee sy eenhede.
Determine the wave number k and give its units.

$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{40} = 0,157 \text{ rad/cm} \quad (2)$$

- (ii) Bepaal die hoeksnelheid ω en gee sy eenhede.
Determine the angular velocity ω and give its units.

$$\omega = 2\pi f = 2\pi(8) = 50,3 \text{ rad/s} \quad (2)$$

- (iii) Bepaal die periode T .
Determine the period T .

$$T = \frac{1}{f} = \frac{1}{8} = 0,125 \text{ s} \quad (2)$$

- (iv) Bepaal die spoed v van die golf.
Determine the speed v of the wave.

$$v = \lambda f = 40(8) = 320 \text{ cm/s} \quad (2)$$

- (v) Bepaal die fasekonstante δ .
Determine the phase constant δ .

$$y = A \sin(kx - \omega t + \delta)$$

$$x=0, t=0: y = A \sin \delta$$

$$15 = 15 \sin \delta$$

$$\therefore \sin \delta = 1$$

$$\therefore \delta = \pi/2 \quad (3)$$

- (vi) Met die waardes soos hierbo bereken, herskryf nou die golf funksie $y = A \sin(kx - \omega t + \delta)$ sodat die fasehoek δ nie daarin voorkom nie.
With the values as calculated above, rewrite the wave function $y = A \sin(kx - \omega t + \delta)$, but in such a way that δ does not appear in the equation.

$$y = A \sin(kx - \omega t + \delta)$$

$$= 15 \sin(0,157x - 50,3t + \pi/2)$$

$$= 15 \cos(0,157x - 50,3t) \quad (3)$$

- (vii) Bereken die waarde van y (in cm) by $x = 3 \text{ cm}$ en $t = 2 \text{ s}$.
Determine the value of y (in cm) at $x = 3 \text{ cm}$ and $t = 2 \text{ s}$.

$$y = 15 \cos[0,157(3) - 50,3(2)]$$

$$= 13,8 \text{ cm} \Rightarrow \quad (2)$$

y direction (at a fixed value of x) is 154,5 cm/s.

$$y = A \cos(kx - \omega t)$$

$$v_y = \frac{dy}{dt} \bigg|_{x=c} = \omega A \sin(kx - \omega t)$$

$$\therefore v_{y, \max} = \omega A \quad \checkmark$$

$$= 50,3 (15)$$

$$= 754,5 \text{ cm/s} \Rightarrow$$

(2)

VRAAG / QUESTION 2

- (a) 'n Voorwerp beweeg in 'n reguit lyn met 'n konstante versnelling a.
An object is moving in a straight line and is experiencing a constant acceleration a.

- (i) Toon deur middel van integrasie aan dat (vir $t_0 \neq 0$) is $v = v_0 + a(t - t_0)$
With integration, show that (with $t_0 \neq 0$) $v = v_0 + a(t - t_0)$

$$a = \frac{dv}{dt} \quad \checkmark$$

$$\int_{v_0}^v dv = \int_{t_0}^t a dt$$

$$[v]_{v_0}^v = a \int_{t_0}^t dt$$

$$v - v_0 = a[t - t_0]$$

$$\text{of } v = v_0 + a(t - t_0)$$

(4)

- (ii) Deur middel van integrasie, toon ook aan dat
With integration, also show that

$$(x - x_0) = v_0(t - t_0) + \frac{1}{2} a(t - t_0)^2$$

$$v = \frac{dx}{dt} \quad dx = v dt$$

$$\int_{x_0}^x dx = \int_{t_0}^t v dv$$

$$[x]_{x_0}^x = \int_{t_0}^t (v_0 + at - at_0) dt$$

$$[x]_{x_0}^x = [v_0 t + \frac{1}{2} at^2 - at_0 t]_{t_0}^t$$

$$x - x_0 = (v_0 t + \frac{1}{2} at^2 - at_0 t) - (v_0 t_0 + \frac{1}{2} at_0^2 - at_0^2)$$

$$= v_0(t - t_0) + \frac{1}{2} at^2 - at_0 t + \frac{1}{2} at_0^2$$

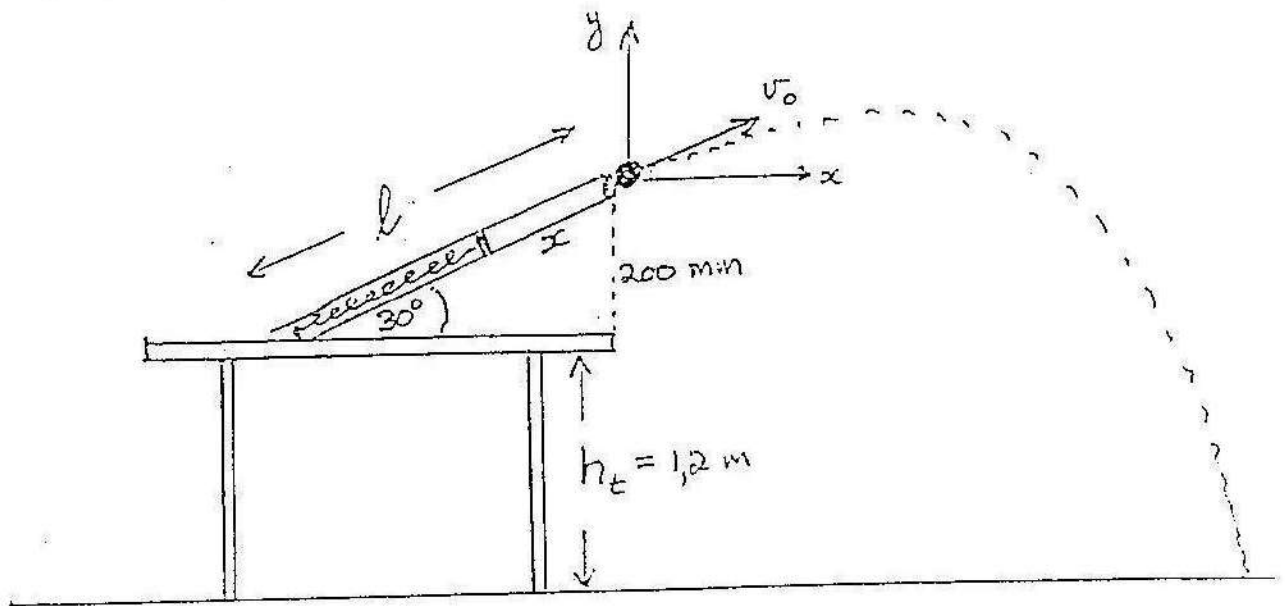
$$= v_0(t - t_0) + \frac{1}{2} a \{t^2 - 2tt_0 + t_0^2\}$$

(5)

Vraag / Question 2(b) (Vervolg / Continue) ...

$$x - x_0 = v_0(t - t_0) + \frac{1}{2} a(t - t_0)^2$$

A marble (mass $m = 500$ gram) is fired from a spring gun (spring constant $k = 155$ N/m) that is mounted at an angle $\theta = 30^\circ$ on a table, as shown in the figure.



- (i) Bereken die lengte ℓ van die geweer (in meter).

Determine the length ℓ of the gun (in meter).

$$\sin \theta = \frac{200}{\ell}$$

$$\ell = \frac{200}{\sin 30^\circ} = 400 \text{ mm} = 0,4 \text{ m} \Rightarrow$$

(1)

- (ii) Die veer word nou sodanig saamgedruk dat sy lengte $\frac{2}{3}$ is van die ontspanne toestand, waar dit dieselfde lengte gehad het as die geweer se loop. Bepaal x (in m).
In the relaxed position the spring has the same length as the barrel. When the spring is compressed its length is $\frac{2}{3}$ that of the barrel. Determine x (in m).

$$x = \frac{1}{3} \ell$$

$$= \frac{1}{3} (0,4) = 0,133 \text{ m} \cdot$$

(1)

- (iii) Bereken die energie E_v wat in die saamgedrukte veer gestoor is.
Determine the energy E_v stored in the compressed spring.

$$E_v = \frac{1}{2} k x^2$$

$$= \frac{1}{2} (155) (0,133)^2$$

$$= 1,37 \text{ J} \Rightarrow$$

(2)

Vraag / Question 2(b)(iv) (Vervolg / Continue) /

verandering van die potensiele energie van die bal $E_p = 0,2 \text{ J}$.

When the spring is released and has moved through the distance x , show that the change in the potential energy of the ball is $E_p = 0,2 \text{ J}$.

$$\begin{aligned} E_p &= m g h \checkmark \\ &= m g (x \sin 30^\circ) \checkmark \\ &= 0,3 (9,8) (0,133 \sin 30^\circ) \\ &= 0,2 \text{ J} \end{aligned}$$

- (v) Toon volledig aan dat die snelheid van die bal, wanneer dit die loop verlaat, is $v_0 = 2,8 \text{ m/s}$.

Show, with the necessary steps, that the velocity of the ball when it leaves the barrel, is $v_0 = 2,8 \text{ m/s}$.

$$\frac{1}{2} k x^2 \checkmark = m g h \checkmark + \frac{1}{2} m v_0^2 \checkmark$$

$$1,37 = 0,2 + \frac{1}{2} (0,3) v_0^2$$

$$v_0^2 = \frac{2(1,37 - 0,2)}{0,3}$$

$$\therefore v_0 = 2,8 \text{ m/s}$$

- (vi) Met die oorsprong van die (x, y) -assessels by die uitgang van die loop (sien figuur), bereken die maksimum hoogte wat die bal totkant die uitgang kan bereik.

With the origin of the (x, y) axis at the exit point of the barrel (see figure), calculate the maximum height of the ball above the exit point.

$$y: v_t^2 = v_0^2 + 2as \checkmark$$

$$0 = (v_0 \sin 30^\circ)^2 - 2(9,8)s$$

$$s = \frac{(2,8 \sin 30^\circ)^2}{2(9,8)} \checkmark = 0,1 \text{ m} \Rightarrow$$

- (vii) Met $y - y_0 = v_{0y} t + \frac{1}{2} a t^2$, bereken die tyd wat die bal sal neem voordat dit die vloer tref.

With $y - y_0 = v_{0y} t + \frac{1}{2} a t^2$, determine the time for the ball to reach the floor.

$$-1,4 - 0 = (2,8 \sin 30^\circ) t - \frac{1}{2} (9,8) t^2$$

$$4,9 t^2 - 2,8 \sin 30^\circ t - 1,4 = 0$$

$$4,9 t^2 - 1,4 t - 1,4 = 0$$

$$t = \frac{-(-1,4) \pm \sqrt{(1,4)^2 - 4(4,9)(-1,4)}}{2(4,9)}$$

$$= 0,69 \text{ s} \Rightarrow$$

Vraag / Question 2(b)(viii)(Vervolg / Continue)

(viii) Indien die afstand vanaf die geweer na die oorkantste muur 2 m is, bereken of die bal die muur of die vloer sal tref. Op tydstep $t_0 = 0$ is die bal by $x_0 = 0$.

With a distance of 2 m to the opposite wall, calculate if the ball will hit the wall or the floor. On time $t_0 = 0$ the ball is at $x_0 = 0$.

$$x - x_0 = v_0 t + \frac{1}{2} a t^2$$

$$x - 0 = (2,8 \cos 30^\circ)(0,69) + 0$$

$$x = 1,67 \text{ m} \checkmark$$

Vloer tref $\checkmark$

(4)

VRAAG / QUESTION 3

- (a) Definieer 'n hoek θ , gemeet in radiale.

Define an angle θ , measured in radians.

$$\theta = \frac{s}{r} \frac{(\text{lengte})}{(\text{straal})}$$

(1)

- (b) Definieer hoeksnelheid ω en gee sy eenhede.

Define angular velocity ω and give its units.

$$\omega = \frac{d\theta}{dt} \checkmark \quad \text{rad/s} \checkmark$$

(2)

- (c) Herlei 'n uitdrukking wat die verband gee tussen lineêre snelheid v en hoeksnelheid ω .

Deduce an equation which gives the relation between linear velocity v and angular velocity ω .

$$\omega = \frac{d\theta}{dt}$$

$$= \frac{d}{dt} \left(\frac{s}{r} \right) \checkmark$$

$$= \frac{1}{r} \left(\frac{ds}{dt} \right) \checkmark$$

$$= \frac{1}{r} v$$

$$v = \omega r \checkmark$$

(3)

- (d) 'n Voorwerp ondergaan 'n konstante hoekversnelling α . Definieer hoekversnelling α .

An object has a constant angular acceleration α . Define angular acceleration α .

$$\alpha = \frac{d\omega}{dt}$$

(1)

Vraag / Question 3(c) (Vervolg / Continue)

Show with integration that:

$$\omega = \omega_0 + \alpha t$$

$$d\omega = \alpha dt$$

$$\int_{\omega_0}^{\omega} d\omega = \alpha \int_{t_0=0}^t dt$$

$$\omega - \omega_0 = \alpha(t)$$

$$\omega = \omega_0 + \alpha t$$

(2)

- (f) 'n Liggaam, met massa m , is opgedeel in massadeeltjies Δm_i . Die liggaam roteer met hoeksnelheid ω . Die lineêre snelheid van die deeltjie Δm_i is v_i en dié deeltjie is op afstand R_i vanaf die rotasie-as. Toon aan dat die rotasie kinetiese energie van die liggaam gegee word deur $K = \frac{1}{2} I \omega^2$, en gee ook 'n uitdrukking vir die traagheids-moment I .

An object with mass m is divided in small mass particles Δm_i . The object is rotating with angular velocity ω . The linear velocity of particle Δm_i , which is at distance R_i from the axis of rotation, is v_i . Show that the rotational kinetic energy of the object is given by $K = \frac{1}{2} I \omega^2$, and also give an expression for the moment of inertia I .

$$K = \sum K_i = \frac{1}{2} \sum \Delta m_i v_i^2 \checkmark$$

$$= \frac{1}{2} \sum \Delta m_i (\omega R_i)^2$$

$$= \frac{1}{2} \sum (\Delta m_i R_i^2) \omega^2 \checkmark$$

$$= \frac{1}{2} I \omega^2$$

$$\text{met } I = \sum \Delta m_i R_i^2 = m R^2 \checkmark$$

(4)

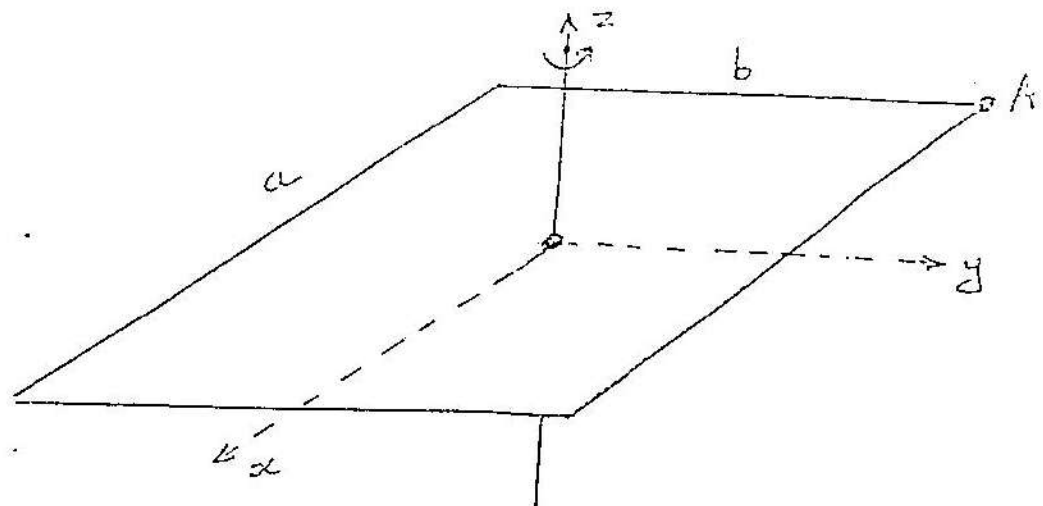
- (g) Die traagheidsmoment om 'n as deur die swaartepunt (CM) van 'n liggaam met massa M is I_{CM} . Skryf 'n uitdrukking neer vir die traagheidsmoment I van die liggaam om enige as wat parallel is aan die as deur die swaartepunt, maar op 'n afstand d daarvan.
The moment of inertia about an axis through the centre of mass (CM) of an object with mass M is I_{CM} . Give an expression for the moment of inertia I about any axis which is parallel to the one through the CM, but a distance d from it.

$$I = I_{CM} + M d^2$$

(1)

- (h) 'n Plat plaat met sylengtes a en b roteer om 'n as deur sy swaartepunt. Toon aan dat die traagheidsmoment om 'n as deur die swaartepunt gegee word deur $I_{CM} = \frac{1}{12} M(a^2 + b^2)$.
A flat plate with sides a and b rotates about an axis through its center of mass. Show that the moment of inertia through the centre of mass is given by $I_{CM} = \frac{1}{12} M(a^2 + b^2)$.

Vraag / Question 3(h) (Vervolg / Continue) ...



$$\sigma = \frac{M}{ab} \checkmark = \frac{dm}{dx dy} \checkmark$$

$$\therefore dm = \frac{M}{ab} dx dy$$

$$\text{en } r^2 = x^2 + y^2 \checkmark$$

$$I = \int r^2 dm \checkmark$$

$$= \iint (x^2 + y^2) \left\{ \frac{M}{ab} dx dy \right\}$$

$$= \frac{M}{ab} \left[\int_{-a/2}^{a/2} x^2 dx \int_{-b/2}^{b/2} dy + \int_{-a/2}^{a/2} dx \int_{-b/2}^{b/2} y^2 dy \right]$$

$$= \frac{M}{ab} \left\{ \left[\frac{1}{3} x^3 \right]_{-a/2}^{a/2} \left[y \right]_{-b/2}^{b/2} + \left[x \right]_{-a/2}^{a/2} \left[\frac{1}{3} y^3 \right]_{-b/2}^{b/2} \right\}$$

$$= \frac{M}{ab} \left\{ \left[\frac{1}{3} \left(\frac{a}{2} \right)^3 - \frac{1}{3} \left(-\frac{a}{2} \right)^3 \right] \left[\frac{b}{2} - \left(-\frac{b}{2} \right) \right] + \left[\frac{a}{2} - \left(-\frac{a}{2} \right) \right] \left[\frac{1}{3} \left(\frac{b}{2} \right)^3 - \frac{1}{3} \left(-\frac{b}{2} \right)^3 \right] \right\}$$

$$= \frac{M}{ab} \left\{ \frac{1}{12} a^3 b + a \cdot \frac{1}{12} b^3 \right\} \checkmark$$

$$= \frac{M}{12} (a^2 + b^2) \Rightarrow$$

- (i) Bepaal die traagheidsmoment om 'n as wat deur A gaan en parallel is aan die z-as.
Determine the moment of inertia about an axis through A and which is parallel to the z-axis.

$$I = I_{cm} + M d^2 \checkmark$$

$$= \frac{1}{12} M (a^2 + b^2) \checkmark + M \left[\left(\frac{a}{2} \right)^2 + \left(\frac{b}{2} \right)^2 \right] \checkmark$$

$$= \frac{1}{12} M (a^2 + b^2) + M \left(\frac{a^2}{4} + \frac{b^2}{4} \right)$$

$$= \left(\frac{1}{12} + \frac{1}{4} \right) M (a^2 + b^2)$$

$$= \frac{1}{3} M (a^2 + b^2) \Rightarrow \checkmark$$

- (a) 'n Voorwerp beweeg langs die x-as. Op tydstip t_1 het dit 'n snelheid v_1 , waarna dit versnel om op tydstip t_2 'n snelheid v_2 te behaal. Gee 'n uitdrukking vir die gemiddelde versnelling $\langle v_x \rangle$ langs die x-as.

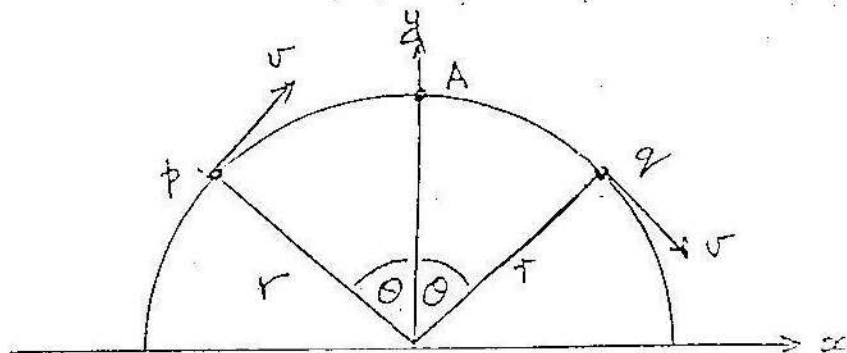
An object moves along the x-axis. It accelerates along the x-axis from velocity v_1 (at t_1) to v_2 (at t_2). Give an expression for the average acceleration $\langle v_x \rangle$ along the x-axis.

$$\langle a_x \rangle = \frac{v_2 - v_1}{t_2 - t_1}$$

(1)

- (b) 'n Voorwerp beweeg met 'n konstante spoed v in 'n sirkelbaan, soos in die figuur.

An object moves with a constant speed v in a circle, as shown in the figure.



- (i) Bereken die gemiddelde versnelling in die x-as rigting.
Calculate the average acceleration in the x-direction.

$$\langle a_x \rangle = \frac{v_{qx} - v_{px}}{t_2 - t_1} \quad \checkmark$$

$$= \frac{v \cos \theta - v \cos \theta}{\Delta t} \quad \checkmark$$

$$= 0 \Rightarrow \quad \checkmark$$

$v_{qx} =$

(3)

- (ii) Toon aan dat die gemiddelde versnelling in die y-as rigting gegee word deur:
Show that the average acceleration in the direction of the y-axis is given by:

$$\langle a_y \rangle = -\frac{v^2}{r} \frac{\sin \theta}{\theta}$$

$$\langle a_y \rangle = \frac{v_{qy} - v_{py}}{\Delta t} = \frac{-v \sin \theta - v \sin \theta}{\Delta t} \quad \checkmark \quad \checkmark$$

$$= -\frac{2v \sin \theta}{\Delta t}$$

$$\text{Maar: } \Delta t = \frac{r(2\theta)}{v} \quad \checkmark$$

$$\therefore \langle a_y \rangle = \frac{-2v \sin \theta}{r(2\theta)/v} = -\frac{v^2}{r} \frac{\sin \theta}{\theta}$$

(3)

- (iii) Toon aan dat die grootte van die versnelling by A gegee word deur $|\vec{a}| = \frac{v^2}{r}$.

Show that the magnitude of the acceleration at A is given by $|\vec{a}| = \frac{v^2}{r}$.

$$\theta \rightarrow 0 \checkmark, \quad \Delta \theta \approx 0 \checkmark$$

$$\langle a_y \rangle = -\frac{v^2}{r} \frac{\theta}{\theta} = -\frac{v^2}{r} \quad (2)$$

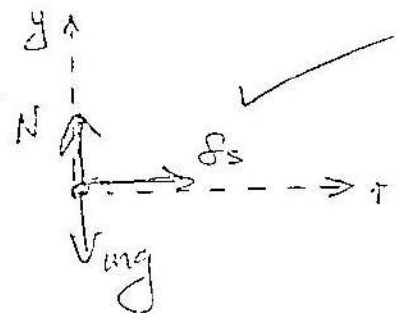
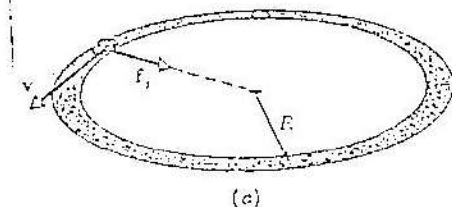
- (c) Die statiese wrywingskoeffisiënt tussen twee vlakke is μ_s . Skryf 'n uitdrukking neer vir die statiese wrywingskrag f_s .

The coefficient of static friction between two surfaces is μ_s . Write an expression for the static friction force f_s .

$$f_s = \mu_s N \quad (1)$$

- (d) 'n Voertuig met massa $m = 1600$ kg beweeg met 'n konstante spoed $v = 20$ m/s op 'n plat sirkelvormige pad met straal $R = 190$ m. Indien die voertuig nie gly nie, toon aan dat $\mu_s = 0,21$. (Toon al die kragte wat op die voertuig inwerk in fig B aan).

A car of mass $m = 1600$ kg is travelling at a constant speed $v = 20$ m/s along a flat, circular road of radius $R = 190$ m. If the car does not slip, show that $\mu_s = 0,21$. (Show all the forces acting on the car in fig. B).



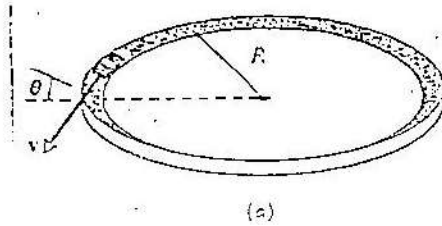
$$f_s = ma = m \frac{v^2}{r} \checkmark$$

$$\therefore \mu_s N v = m \frac{v^2}{r}$$

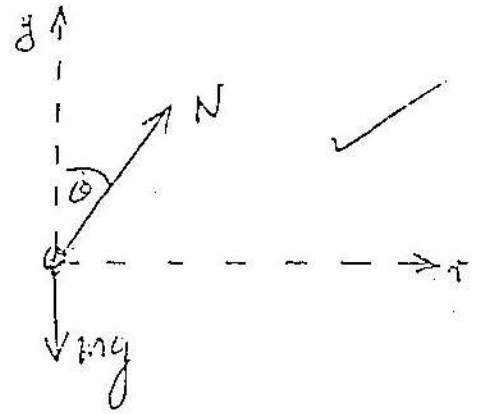
$$\mu_s (mg) = m \frac{v^2}{r}$$

$$\mu_s = \frac{v^2}{gR} = \frac{(20)^2}{9,8(190)} = 0,21 \checkmark$$

- (e) Die voertuig in vraag (d) beland op die sirkelvormige pad met 'n yslaag op, sodat daar geen wrywing is nie. Wat is die hoek θ wat die pad moet hê sodat die voertuig nie afgly nie? (Toon al die kragte wat op die voertuig inwerk in fig. B aan.)
The car in question (d) hits the circular road on which there is a layer of ice – and therefore no friction. What is the angle θ of the banked road in order that the car does not slip? (Show all the forces acting on the car in fig. B).



(b)



$$N_y = N \cos \theta - mg = 0$$

$$N \cos \theta = mg \quad \checkmark$$

$$N_x = N \sin \theta = \frac{mv^2}{R} \quad \checkmark$$

$$\frac{N \sin \theta}{N \cos \theta} = \frac{mv^2/R}{mg} = \frac{v^2}{gR}$$

$$\tan \theta = \frac{v^2}{gR} \quad \checkmark$$

$$= \frac{(20)^2}{9.8 (190)}$$

$$\theta = 12^\circ \Rightarrow \checkmark$$

(5)

- (a) 'n Voorwerp met massa m beweeg met 'n snelheid $\vec{v}$. Gee 'n uitdrukking vir die momentum van die voorwerp.
An object with mass m is moving with velocity $\vec{v}$. Give an expression for the momentum of the object.

$$\vec{p} = m\vec{v}$$

(1)

- (b) Wat is die voorwaarde(s) vir 'n nie-elastiese botsing.
Under what condition(s) will a collision be inelastic?

$$\vec{p}_{\text{voor}} = \vec{p}_{\text{na}}$$

✓✓

(2)

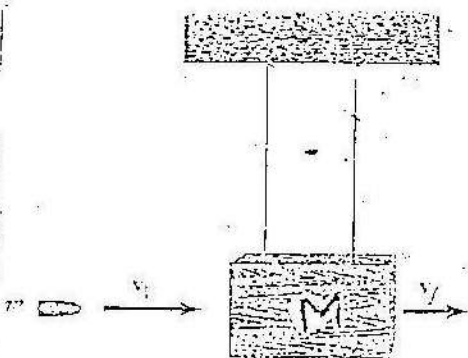
- (c) (i) 'n Koeël met massa m en snelheid v_1 word in 'n blok hout (met massa M) ingeskiet, en sit daarna vas in die blok. Die blok hang aan twee ligte toue, en onmiddellik nadat die koeël in die blok in is beweeg hulle met snelheid v_f . Bereken v_f indien $m = 5$ gram, $M = 1$ kg en $v_1 = 199$ m/s.
A bullet with mass m and velocity v_1 is fired into a block of wood (mass M) suspended from some light wires. The bullet embeds in the block, and the system then has a velocity v_f . Calculate v_f if $m = 5$ gram, $M = 1$ kg and $v_1 = 199$ m/s.

$$m v_1 = (m + M) v_f$$

$$v_f = \frac{m v_1}{(m + M)}$$

$$= \frac{(0,005) 199}{1,005}$$

$$= 0,99 \text{ m/s} \Rightarrow$$



(2)

- (ii) Bepaal die kinetiese energie E_i van die koeël.
Determine the kinetic energy E_i of the bullet.

$$E_i = \frac{1}{2} m v_1^2 = \frac{1}{2} (0,005) (199)^2$$

$$= 99 \text{ J} \Rightarrow$$

(2)

- (iii) Bepaal die kinetiese energie E_f nadat die koeël in die blok in is en hulle met snelheid v_f beweeg.
Determine the kinetic energy E_f of the bullet and block moving with velocity v_f

$$E_f = \frac{1}{2} (m + M) v_f^2$$

$$= \frac{1}{2} (1,005) (0,99)^2$$

$$= 0,49 \text{ J} \Rightarrow$$

(2)

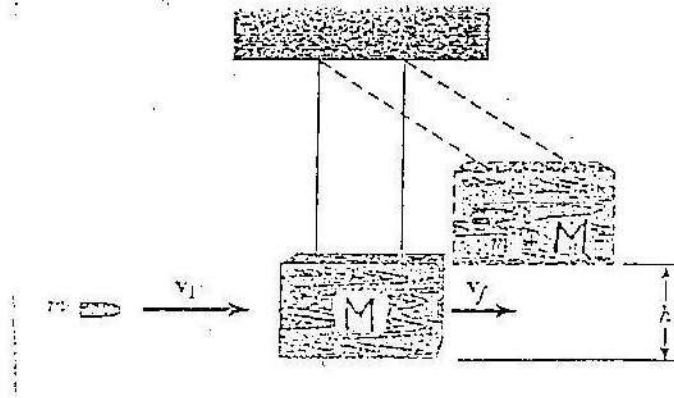
Vraag / Question 5(c)(iv) (vervolg / continue) / ...

- (iv) Was dit 'n elastiese of nie-elastiese botsing?
Was the collision elastic or inelastic?

Nie-elasties.

(1)

- (v) Die blok swaai op deur 'n hoogte h . Bereken h .
The block swings upwards through a height h . Determine h .



$$mgh = E_f$$

$$h = E_f / mg = \frac{0,49}{(0,005) 9,8}$$

$$= 0,05 \text{ m} \Rightarrow$$

(2)